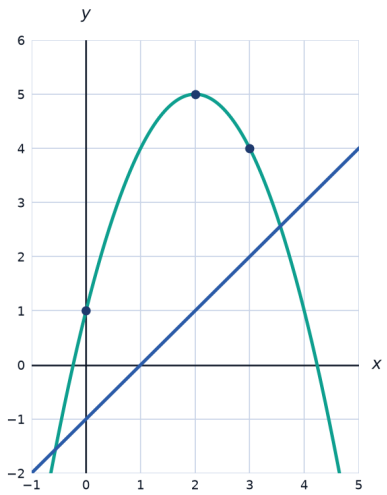
Sequences & Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Arithmetic series

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(5)$.

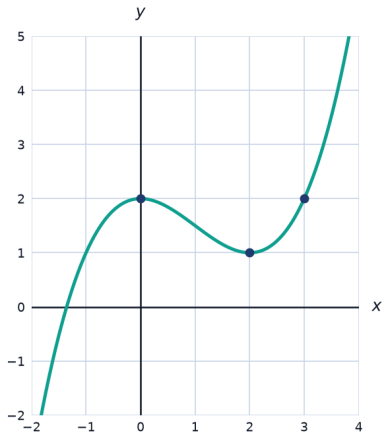


2 [Foundational] • Target: Arithmetic series

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

3 [Foundational] • Target: Geometric sequences

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



4 [Foundational] • Target: Geometric sequences

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

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5 [Foundational] • Target: Geometric sequences

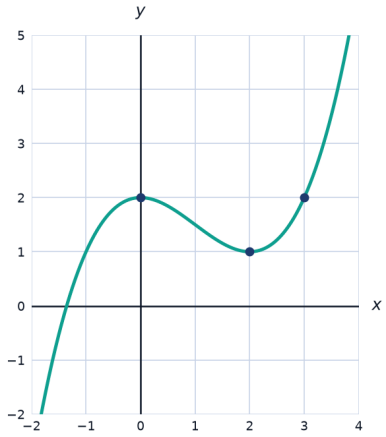
On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

6 [Foundational] • Target: Geometric sequences

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Geometric sequences

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Geometric sequences

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

9 [Intermediate] • Target: Geometric sequences

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

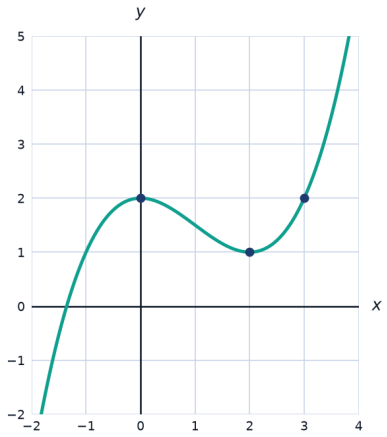
10 [Intermediate] • Target: Geometric sequences

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

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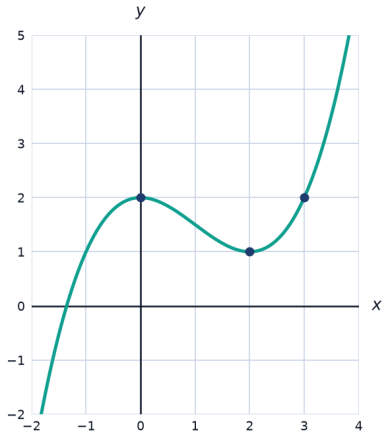
11 **[Intermediate]** • **Target: Geometric sequences**

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



12 **[Intermediate]** • **Target: Geometric sequences**

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 **[Intermediate]** • **Target: Geometric sequences**

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

14 **[Intermediate]** • **Target: Geometric sequences**

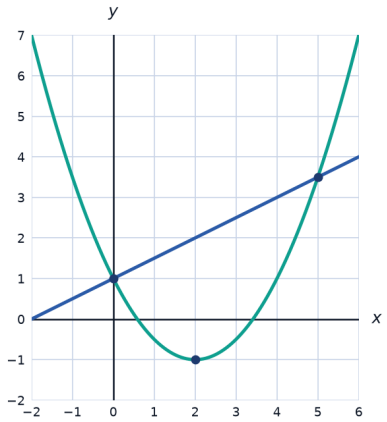
On the coordinate plane, a line with slope $m = 5$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Geometric series • Cross-Lesson: 11.01 Probability & Statistics

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



16 [Challenging] • Target: Geometric series • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

17 [Challenging] • Target: Geometric series • Cross-Lesson: 3.01 Polynomial Functions

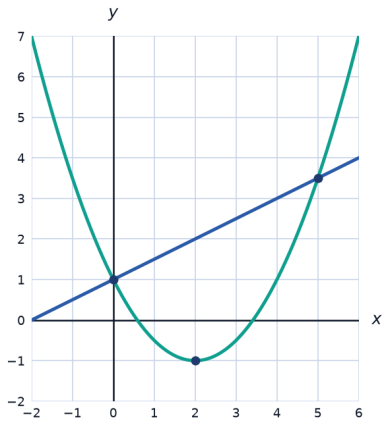
On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

18 [Challenging] • Target: Geometric series • Cross-Lesson: 4.01 Rational Expressions

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Geometric series • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.



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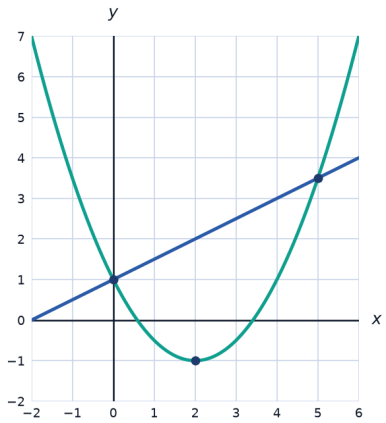
20 [Challenging] • Target: Geometric series • Cross-Lesson: 6.01 Radicals & Rational Exponents

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(2)$.



a. 19

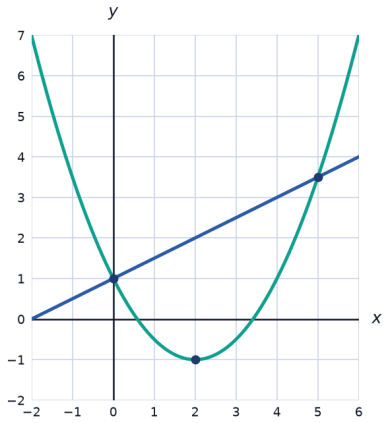
b. 24

c. 10

d. 23

22 [Foundational] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

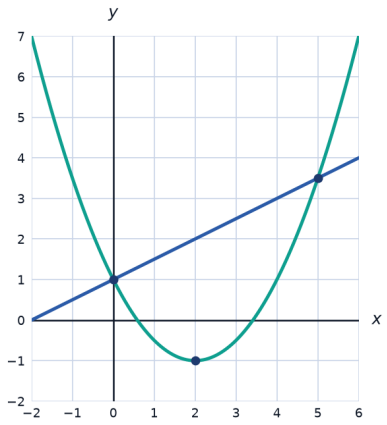
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(5)$.



a. 34

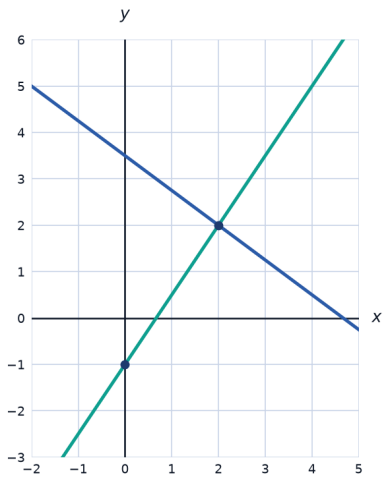
b. 39

c. 25

d. 38

24 [Foundational] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(2)$.



a. 15

b. 21

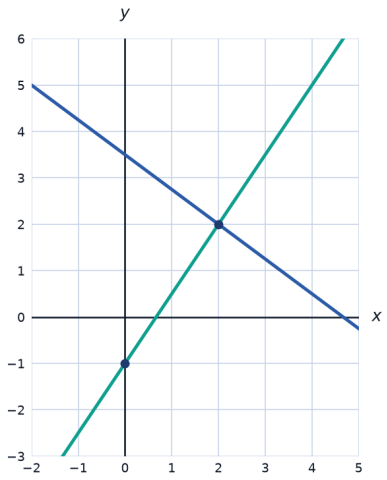
c. 12

d. 19

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25 [Foundational] • Target: Arithmetics sequences

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

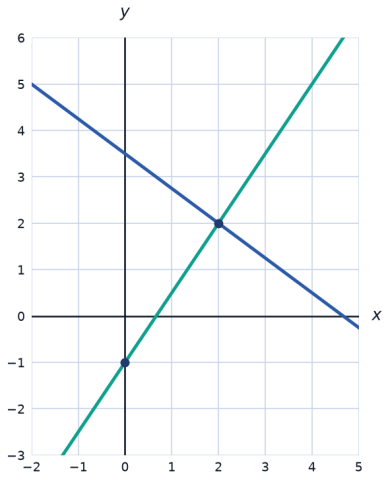
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(5)$.



a. 33

b. 39

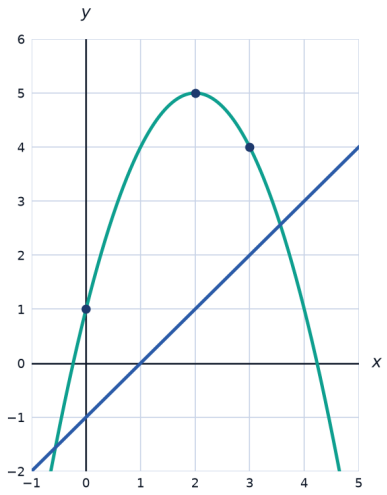
c. 30

d. 37

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27 [Foundational] • Target: Arithmetic series

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(2)$.



a. 8

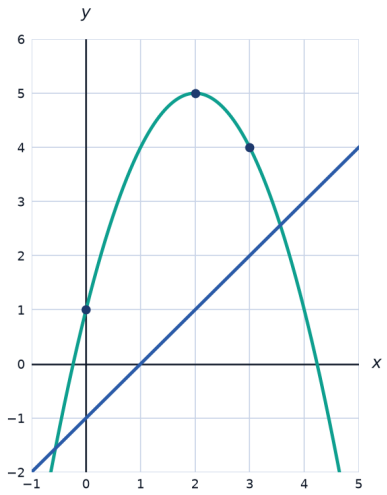
b. 10

c. 4

d. 12

28 [Foundational] • Target: Arithmetic series

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Notation for sequences and series

A data set recorded for notation for sequences and series consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

31 [Intermediate] • Target: Notation for sequences and series

For the model $f(x) = 5x + 9$ in notation for sequences and series, evaluate $f(4)$.

a. 29

b. 34

c. 20

d. 33

32 [Intermediate] • Target: Notation for sequences and series

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

33 [Intermediate] • Target: Arithmetics sequences

A data set recorded for arithmetics sequences consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

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34 [Intermediate] • Target: Arithmetics sequences

For the model $f(x) = 6x + 3$ in arithmetics sequences, evaluate $f(4)$.

a. 27

b. 33

c. 24

d. 31

35 [Intermediate] • Target: Arithmetics sequences

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

36 [Intermediate] • Target: Arithmetic series

A data set recorded for arithmetic series consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

37 [Intermediate] • Target: Arithmetic series

For the model $f(x) = 2x + 4$ in arithmetic series, evaluate $f(4)$.

a. 12

b. 14

c. 8

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 4.01 Rational Expressions

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

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39 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97°

b. 107°

c. 89°

d. 149°

40 [Challenging] • Target: Geometric sequences • Cross-Lesson: 6.01 Radicals & Rational Exponents

In ΔABC studying geometric sequences, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 149°

41 [Challenging] • Target: Geometric sequences • Cross-Lesson: 8.01 Conic Sections

A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

42 [Challenging] • Target: Geometric series • Cross-Lesson: 9.01 Trigonometric Functions

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

43 [Challenging] • Target: Geometric series • Cross-Lesson: 1.01 Functions & Their Graphs

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 148°

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44 [Challenging] • Target: Geometric series • Cross-Lesson: 10.01 Matrices & Systems

In ΔABC studying geometric series, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89°

b. 99°

c. 81°

d. 148°

45 [Challenging] • Target: Geometric series • Cross-Lesson: 11.01 Probability & Statistics

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

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